Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Forward deployment engineer with 8+ years embedding with enterprise teams to translate business goals into production AI systems. Built multi-agent agentic architectures end-to-end: **AgentProbe** (3-agent LangGraph runtime, 5-metric LLM-as-judge evaluation engine, multi-provider LLM routing, RAG with vector retrieval -- live at github.com/venkatalakshmiparimala/AgentProbe) and **Leasel** (8-agent HITL pipeline, 93.6% precision across 1,240 enterprise documents). Operationalizes LLMs across cloud infrastructure (AWS, Azure); grounds AI models in live business context via vector retrieval to eliminate AI hallucination on domain data. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Multi-Agent Agentic Architecture (LangGraph, LangChain) | LLM Evaluation & Model Assessment | RAG Pipelines & Vector Search (ChromaDB) | Generative AI Deployment at Scale | Python · SQL · TypeScript | AWS (S3, EC2, RDS, Glue) · Azure | Enterprise Data Pipelines & ETL | Consultative Cross-functional Delivery

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- **Designed and deployed an agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) -- operationalized across marketing and commercial survey workflows, delivering risk-aware compliance decisions with grounded, cited reasons.
- **Built a RAG retrieval layer** (ChromaDB vector engine, FAA Part 107 corpus) grounding every compliance check in source regulation -- eliminating AI hallucination on domain-specific rules. Engineered validation guardrails blocking non-compliant outputs before export.

Entrata -- PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer -- Lehi, UT (Remote)

- **Embedded cross-functionally with compliance, operations, and product teams** to translate high-level business goals into production architectures -- shipped 20+ full-stack modules across 120+ Agile sprints for 100K+ monthly active users; provided transparent status updates and proactive stakeholder communication throughout delivery cycles.
- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) -- LLM-powered automation with output validation, quality gates, and compliance controls; built streaming data workflows replacing manual Excel operations at enterprise scale.
- **Built and optimized enterprise data pipelines** for real-time financial operations (HAP, HUD payments, income eligibility, AMI-based pricing) across 100K+ unit portfolios -- zero federal/state compliance violations; optimized 200+ SQL queries (30% performance gain) via execution plan tuning, indexing, and partitioning.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines -- zero security incidents. Led Docker-based microservices redesign: 50% scalability improvement, 30% deployment time reduction.
- Eliminated 80% of manual operational workload via Python/Bash automation; conducted 300+ peer code reviews; mentored 5+ engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed faculty dashboards improving comprehension 40%. Mentored 600+ students in Python and database design.

PROJECTS

AgentProbe: Multi-Agent Evaluation & LLM Observability Platform Live

3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), tool-use patterns, and inter-agent state handoffs. CI-integrated 5-metric LLM-as-judge evaluation engine (factual accuracy, hallucination, retrieval, completeness, compliance) -- every commit auto-scored, regressions blocked before production. ChromaDB RAG pipeline with trace-level observability: per-run metrics, regression drift, and retrieval quality monitoring.

Python · LangGraph · LangChain · FastAPI · ChromaDB · React · Docker · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Leasel: Enterprise AI Document Intelligence & 8-Agent HITL Pipeline

Dual-mode enterprise AI platform: Document Intelligence (OCR ingestion, vector retrieval, 100% citation correctness, human-review gate) + Support Investigation (8-agent LangGraph pipeline with conditional HITL escalation when confidence is too low). 93.6% precision / 87.9% F1 across 10 compliance rules on 1,240 enterprise documents (5 US states + federal).

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Reduced manual clinical retrieval time ~60% by building an NLP platform (LLMs, BERT, embeddings) translating natural language queries (EN/ES) to FHIR APIs with HIPAA-compliant RBAC. Deployed on cloud infrastructure with Docker containerization.

Flask · React · LLMs · FHIR · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, R, Bash

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Agentic Architecture, HITL Design, LLM Evaluation, LLM-as-Judge, Prompt Engineering, RAG Pipelines, Vector Search, ChromaDB, Embeddings, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, GitHub Actions CI/CD, PostgreSQL, MySQL, ETL, Data Pipelines, Microservices, RabbitMQ